

Integração de Funções Racionais de Seno e Cosseno.

Quando a integral tem a forma $\int R(\cos x, \sin x) dx$, isto é, o integrando é uma função racional de seno e cosseno, essa integral, pode ser transformada numa integral de uma função racional em outra variável, digamos t .

Para isso, devemos fazer a substituição.

$$t = \tan\left(\frac{x}{2}\right) = t \quad \text{com} \quad -\pi < x < \pi$$

Para isso, precisamos determinar $\sin x$, $\cos x$ e dx em função da variável t .

$$\sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2} \quad dx = \frac{2dt}{1+t^2}$$

$$\textcircled{\text{I}} \operatorname{sen} x$$

$$\operatorname{sen} x = \operatorname{sen} \left(\frac{x}{2} + \frac{x}{2} \right)$$

$$\operatorname{sen} x = \frac{2 \operatorname{sen} \left(\frac{x}{2} \right) \cdot \cos \left(\frac{x}{2} \right)}{1}$$

$$\operatorname{sen} x = \frac{2 \operatorname{sen} \left(\frac{x}{2} \right) \cdot \cos \left(\frac{x}{2} \right)}{\operatorname{sen}^2 x + \cos^2 x}$$

$$\operatorname{sen} x = \frac{2 \operatorname{sen} \left(\frac{x}{2} \right) \cdot \cos \left(\frac{x}{2} \right)}{\cos^2(x) \cdot \frac{\operatorname{sen}^2 \left(\frac{x}{2} \right) + \cos^2 \left(\frac{x}{2} \right)}{\cos^2 \left(\frac{x}{2} \right)}}$$

$$\operatorname{sen} x = 2 \cdot \frac{\operatorname{sen} \left(\frac{x}{2} \right)}{\cos \left(\frac{x}{2} \right)} \cdot \frac{\cos \left(\frac{x}{2} \right)}{\frac{\operatorname{sen}^2 \left(\frac{x}{2} \right)}{\cos^2 \left(\frac{x}{2} \right)} + \frac{\cos^2 \left(\frac{x}{2} \right)}{\cos^2 \left(\frac{x}{2} \right)}}$$

$$\operatorname{sen} x = \frac{2 \operatorname{tg} \left(\frac{x}{2} \right)}{\operatorname{tg}^2 \left(\frac{x}{2} \right) + 1}$$

$$\operatorname{sen} x = \frac{2t}{1+t^2}$$

$$\textcircled{\text{II}} \quad \cos x = \cos\left(\frac{x}{2} + \frac{x}{2}\right)$$

$$\cos x = \frac{\cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)}{1}$$

$$\cos x = \frac{\cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)}{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)}$$

$$\cos x = \frac{\frac{\cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)}{\cos^2\left(\frac{x}{2}\right)}}{\frac{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)}{\cos^2\left(\frac{x}{2}\right)}}$$

$$\cos x = \frac{1 - \tan^2\left(\frac{x}{2}\right)}{1 + \tan^2\left(\frac{x}{2}\right)}$$

$$\cos x = \frac{1 - t^2}{1 + t^2}$$

$$\textcircled{\text{III}} \quad \operatorname{tg}\left(\frac{x}{2}\right) = t$$
$$\frac{x}{2} = \operatorname{arctg} t$$

$$x = 2 \operatorname{arctg} t$$

$$\frac{dx}{dt} = 2 \cdot \frac{1}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$\int \frac{dx}{\frac{2t}{1+t^2}}$$